05.06.2024

EE2004 MICROPROCESSORS FINAL EXAM

Name:

CLOSED BOOK, CLOSED NOTES EXAM. NO CALCULATORS OR CHEAT SHEETS ALLOWED.
PLEASE PROVIDE YOUR ANSWERS IN DESIGNATED PLACES.
DURATION: 60 min.

Q1) (10 points) Suppose R# is an 8-bit register. $R1$ and $R2$ are given as follows:

$R_1 = x_7x_6x_5x_4\ x_3x_2x_1x_0$

$R_2 = y_7y_6y_5y_4\ y_3y_2y_1y_0$

Write an assembly code to generate $R3$ from $R1$ and $R2$.

$R_3 = \overline{x_7}\ y_6\ \overline{y_5}\ x_4\ x_3\ \overline{x_2}\ y_1\ \overline{y_0}$

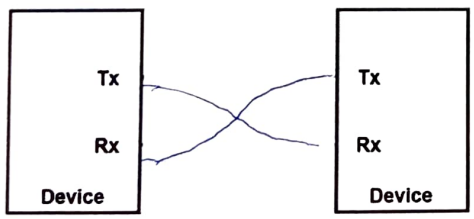
Available instruction set is: {AND, ORR, EOR}.

```
EOR R3, R1, #0b10000100
AND R3, R3, #0b10011100
EOR R4, R2, #0b00100001
AND R4, R4, #0b01100011
ORR R3, R3, R4.
```

Q2) (20 points) For both digital communication protocols below answer following,

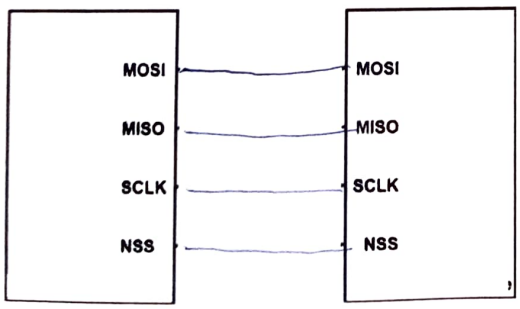
I) Connect pins on the diagram.
II) Give name of the communication protocol.
III) Is it serial or parallel?
IV) Is it synchronous or asynchronous?
V) Is it possible to apply master and slave operation?

a)



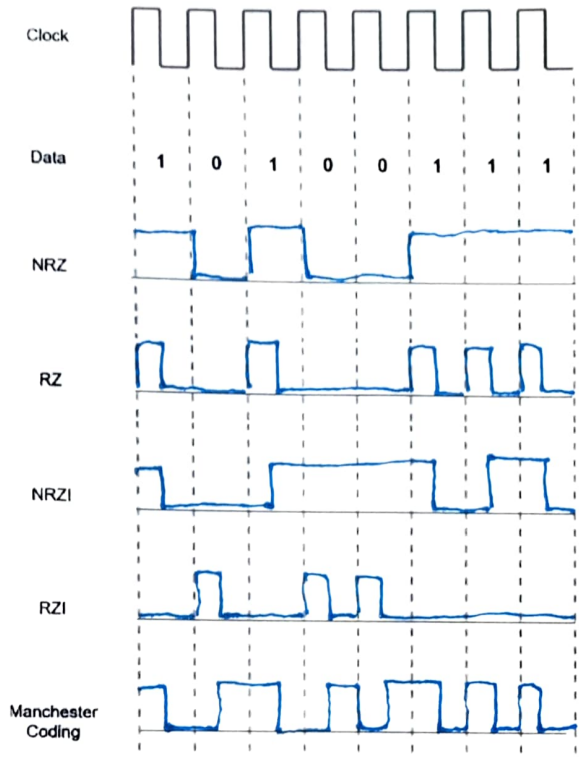
UART (Universal asynchronous receiver transmitter)
Serial
asynchronous
master and slave definitions do not apply

b)



SPI (serial peripheral interface)
Serial
Synchronous
master and slave is used

Q3) (15 points) An 8-bit data packet to be transmitted as 10100111. Sketch data encoding waveforms of non-return to zero, return to zero, non-return to zero inverted, return to zero inverted, and Manchester coding.



Q4) (5 points) A DAC is used to output a sine wave using SysTick Interrupts and a sine-wave table. Assume the DAC has 8 bits, the DAC output is connected to a speaker, the SysTick ISR executes at 100kHz. How many elements should the sine table include in order to produce a 400 Hz sound?

$\frac{100k}{400Hz} = 250$ elements

Q5) (5 points) Three microcontrollers (A, B & C) have maximum clock speeds 10MHz, 20MHz, 24MHz respectively. Microcontroller A divides its clock by 4 to give one machine cycle, microcontroller B by 8, and microcontroller C by 12. Microcontrollers A & C take 2 machine cycles to perform an instruction, while microcontroller B takes three cycles. Place the microcontrollers in order of the speed in which they can perform that instruction?

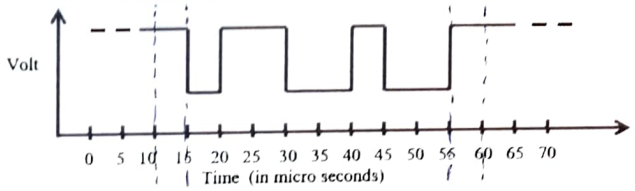
A: $10/4 = 2.5$ MHz 1 cycle $= \frac{1}{2.5} = 0.4\,\mu s$ $\times 2 = 0.8\,\mu s$

B: $20/8 = 2.5$ MHz 1 cycle $= \frac{1}{2.5} = 0.4\,\mu s$ $\times 3 = 1.2\,\mu s$

C: $24/12 = 2$ MHz 1 cycle $= \frac{1}{2} = 0.5\,\mu s$ $\times 2 = 1\,\mu s$

Speed: A > C > B

Q6) (15 points) Reverse-engineer UART parameters from the trace observed at a receiver below.



a) What is the data value transferred over the UART in hexadecimal?

0 0 1 0 0 1 1 0 = 0x 26

or 0 1 1 0 0 1 0 0 = 0x 64

b) What is the baud rate in bits/sec?

$1/5_{\mu s}$ = 200 kbit/s

Q7) (5 points) You have a 12-bit, 0 to 3V range ADC. If the ADC input is 1.25 V, what will be the digital value in hex returned by this ADC?

$\frac{1.25 \times 4096}{3} = 1706 = 0x6AA$

Q8) (15 points) When using the system timer, there is just one callback function named HAL_IncTick(void) for SysTick. This function is called every millisecond.

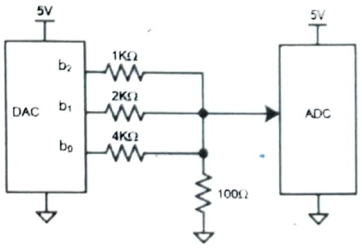
a) Fill question marks (there are two) in the listing below.
b) Explain the operation done by the code.

```
/* USER CODE BEGIN PV */
uint16_t sysTickCnt = 0;
/* USER CODE END PV */

/* USER CODE BEGIN 0 */
void HAL_IncTick(void)
{
sysTickCnt++;          1000
if (sysTickCnt >= ??  )
{
HAL_GPIO_TogglePin(GPIOG, GPIO_PIN_13);
sysTickCnt = ??;
}                      0
}
/* USER CODE END 0 */
```

LED connected to PG13 is toggled at 1s intervals using SysTick.

Q9) (10 points) Given left below the 3-bit DAC with 5V reference, provide the NUMERICAL EXPRESSION for the input voltage seen at the ADC input for the following DAC output codes with the 100 Ω load attached. The ADC reference is also 5V.



	DAC output code (binary) b_2 b_1 b_0	ADC input voltage expression (V)
(a)	000	0 V
(b)	001	$V = \frac{1}{41} \times 5 = \frac{5}{41}$ V ≅ 0.12 V
(c)	010	$V = \frac{1}{21} \times 5 = \frac{5}{21}$ V ≅ 0.24 V
(d)	100	$V = \frac{1}{11} \times 5 = \frac{5}{11}$ V ≅ 0.46 V
(e)	111	$1 // 2 // 4 = 570\Omega$ $V = \frac{10}{67} \times 5 = \frac{50}{67}$ V ≅ 0.75 V

Q10) (15 points) In Listing below, the timer1 is started at the rising edge of the GPIO interrupt generated by the user button press. At the falling edge of the user button press, the timer1 is stopped and elapsed time is measured in milliseconds. If the elapsed time is greater than 3000 ms, then the green LED on the STM32F4 board turns on. Otherwise, the green LED turns off. Finally, the timer1 is reset to measure the next pressing time.
Fill missing lines (five lines are missing) in the code.

```
#include "mbed.h"

InterruptIn userButton(BUTTON1);
DigitalOut greenLED(LED1);
Timer timer1;

uint32_t elapsedMs;

void triggerRise()
{
  timer1.start( );
}

void triggerFall()
{ timer1.stop( );
elapsedMs = chrono::duration_cast<chrono::milliseconds>(timer1.elapsed_time()).count();

if (elapsedMs > 3000)
------greenLED = 1;
else
------greenLED = 0;
--------
}     timer1.reset( );
int main()
{
userButton.rise(&triggerRise);
userButton.fall(&triggerFall);

while (1);
}
```